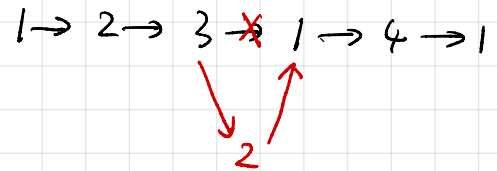


T

	1	2	3	4
1	0	0	<u>1</u>	<u>1</u>
2	1	0	0	0
3	0	1	0	0
4	<u>1</u>	0	0	0

	1	2	3	4
1	0	0	0	1
2	1	0	0	0
3	0	1	0	0
4	1	0	0	0

Action : Add Port.



(1, 3) $T_{13} = 1$

⇒ 1) $T_{13} = 0$

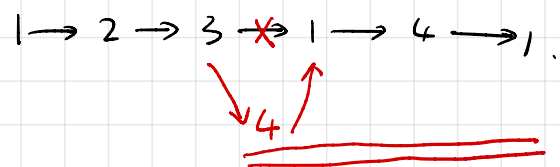
2) $T_{1k} = 1, T_{k3} = 1, k = 2.$

Generally, "add port"

$\Delta T_{ij} = 1, T_{ik} = 0, T_{kj} = 0$

⇒ $T'_{i1} = 0, T'_{ik} = 1, T'_{kj} = 1.$

Assumptions: $T_{ik} = 0, T_{kj} = 0.$



Input → Action Distribution

T_{ij} $\xrightarrow{\text{Softmax}} P_{(i,j,k)} \text{ (Distrib.)}$

$\uparrow M_{(i,j,k)} \text{ (Score)}$

$\uparrow$

$\begin{cases} \text{if } T_{ij} = 0, & M_{ijk} = 0, \forall k, \\ \text{if } T_{ij} = 1, & \begin{cases} \text{if } T_{ik} = 0, T_{kj} = 0, & \underline{M_{ijk} = -\infty} \leftarrow NN \\ \text{otherwise} & M_{ijk} = 0 \end{cases} \end{cases}$

1 → 2 → 3 → 1 → 4

Delete "3"

1 → 2 → 3 → 4 → 1

	1	2	3	4
1	0	1	1	1
2	1	0	0	0
3	0	1	0	0
4	1	0	0	0

$$T_{ij} = 0, T_{kj} = 1, T_{ik} = 1, \quad (i=1, j=2, k=3).$$

$$\Rightarrow T'_{ij} = 1, T_{kj} = 0, T_{ik} = 0,$$

Input → Output

$T(i,j)$

$P(i,j,k)$ { P_{ijk} : Prob. that rm port k .
between i, j .
Ori. $i \rightarrow k \rightarrow j$
New: $i \rightarrow j$.

$M(i,j,k)$

if $T_{ij} = 1$, $M_{ijk} = 0$,

if $T_{ij} = 0$, { if $T_{kj} = 1, T_{ik} = 1, \Rightarrow M_{ijk} = \infty$,
otherwise $M_{ijk} = -\infty$